



**MECHANICAL ENGINEERING
SENIOR DESIGN CAPSTONE
PROPOSAL**

UAV Powertrain Health Monitoring Framework

A Proposal submitted in partial
fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE

in

MECHANICAL ENGINEERING

and

COMPUTER SCIENCE

To the Department of Mechanical and Materials Engineering,
Computer Science,
College of Engineering and Applied Science at the University of Cincinnati.

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Executive Summary

The goal of this project is to develop a UAV powertrain health monitoring framework that integrates hardware based acoustic sensing with machine learning driven diagnostics. The framework will address two UAV subsystems: motors and batteries, which are among the most failure prone components in UAV operations.

The project involves building a UAV test platform equipped with USB microphones, current/voltage sensors, and a drone stand for controlled, repeatable, and safe testing. Microphone placement has been selected to balance signal to noise ratio so that both motor and propeller related faults can be captured. Data will be recorded in two ways: in a stand and while in flight to validate motor performance under realistic conditions. Multiple methods of recording motor and propeller sound will be designed and compared, along with a strain measuring system that will advise the health monitoring framework regarding structural stresses.

Two machine learning models will be developed and integrated: a 1D CNN Autoencoder for semi supervised detection of motor and propeller anomalies from acoustic data, and a DeepONet for supervised battery degradation modeling, RUL prediction, and flight time estimation. Both models will be optimized for deployment on onboard computer hardware with a NVIDIA Jetson Nano used for in-flight computing.

The final system will offer anomaly detection and diagnostics while in flight. This project and the associated research directly enable integration into IASMS by combining deep learning, mechanical design, and hardware sensing. This will enhance UAV safety and reliability for all applications.

Nomenclature

Due to the extensive use of acronyms used, below is a summary of all unclear acronyms and terms included in this proposal.

1D CNN: One Dimensional Convolutional Neural Networks

DeepONet: Deep Operator Network

I2S: serial communication protocol designed for transmitting two-channel digital audio as pulse code modulation between integrated circuit components of an electronic device

IASMS: In-time Aviation Safety Management System

LSTMs: Long Short Term Memory

MEMS: Micro Electrical Mechanical Systems

RUL: Remaining Useful Life

SoC: State of Charge

SQL: Structured Query Language

UAV: Unmanned Aerial Vehicles

Introduction

UAVs, particularly quadcopters and octocopters, are increasingly deployed in applications such as agriculture, property mapping, and rescue operations [1]. Due to the potential danger of these situations, it is vital that UAVs operate properly, and issues are identified immediately. However, UAV safety and reliability are constrained by the susceptibility of their powertrain components,

such as motors and batteries, to degradation and failure. Failures in either system can lower performance, reduce safety, and require costly repairs.

To address the problem of motor degradation, the project will leverage acoustic monitoring to diagnose motor health. Motors produce unique sound profiles when faults are present [2]. By integrating microphones close to the motors, these profiles can be captured and analyzed. Mechanical design decisions such as method of recording audio, placement relative to motors, and vibration isolation directly affect data quality and reliability. Data will be collected using both stand-mounted experiments for repeatable data and in-flight tests for realistic data.

This project addresses the issue of lithium-ion battery degradation by implementing a data driven battery health model. Voltage and current cycling data will be recorded, and electrical sensors will be integrated into the UAV platform. Additionally, the temperature of the battery may be approximated using a sensor such as a thermocouple. This will enable predictive analytics for degradation trajectories, RUL estimation, and remaining flight time.

The analytics layer complements the hardware, translating raw signals into applicable insights. Motor health will be monitored with a 1D CNN Autoencoder trained on healthy acoustic data to detect anomalies semi-supervised, without needing large and detailed datasets of faults. Battery health will be modeled with a DeepONet, trained on cycling data to predict RUL and endurance. Both models will be deployed on the UAV onboard computing hardware, supporting real time monitoring.

This project originated from a collaborative research effort involving Dr. Chao, Dr. Kulkarni from NASA, and Dr. Kumar at the University of Cincinnati. Prishha Chawla conducted a feasibility study during a study abroad program at the Zurich University of Applied Sciences under Dr. Chao, investigating whether specific acoustic signatures could serve as reliable indicators of UAV motor faults, with Dr. Kulkarni exploring the potential integration of this approach into NASA's Fit-2-Fly pre-flight framework for octocopters. Following promising results, the research expanded to UAV battery systems under Dr. Kumar, where Prishha applied machine learning models to predict battery voltage using online datasets, with Dr. Chao guiding the machine learning work and Dr. Kumar providing mechanical expertise. After confirming feasibility for both motors and batteries, the project progressed to real-world implementation, integrating acoustic and battery models into an onboard UAV system for in-flight testing and prediction of remaining flight time. This effort evolved into the current project, assembling a multidisciplinary team with expertise in mechanical design, machine learning, and software integration with Dr. Chao, Dr. Kulkarni, Dr. Kumar, with Dr. Kumar's PhD student Luke Busse serving as advisors. This project responds directly to a defined problem identified by the collaborators, addressing a known need in UAV health monitoring rather than proposing a solution to an unknown issue.

This proposal contains the following sections: qualifications of the team, background and justification of the project, a statement of work to be completed, the schedule and management plan, cost breakdown, signatures from the team, works cited, and appendices.

Qualifications

This project requires a multidisciplinary team able to address challenges in mechanical design, hardware integration, data collection, and machine learning. The team will leverage experience in mechanical engineering, computer science, and data science. Complete skills of the team are outlined below.

Ally Blair – Mechanical Engineering

Leads the physical UAV build, sensor selection, design, and placement, vibration isolation, and test stand design for safe and repeatable experiments.

- Experienced in mechanical design, system integration, and experimental testing.
- Proficient in 3D modeling using SolidWorks, Siemens NX, and Autodesk Inventor.
- Designed, assembled, tested, and verified electromechanical and pneumatic systems.
- Experience developing custom fixtures and test systems for pneumatic testing per customer guidelines.

Prissha Chawla – Computer Science and Statistics

Leads development of the 1D CNN Autoencoder for motor anomaly detection and contributes to data-driven battery health modeling

- Expert in machine learning, predictive modeling, and data science.
- Proficient in Python, R, SQL, and deep learning frameworks including CNNs, LSTMs, and DeepONet.
- Certified in analytics and cloud computing, including AWS, Tableau, and Alteryx.

Siddharth Urankar – Computer Science with Mathematics Minor, Business Analytics Certificate, Software Engineering Certificate

Leads development of the DeepONet framework for battery health modeling and contributes to model optimization and deployment on UAV hardware.

- Experienced in AI, computer vision, and software engineering via research at UC Digital Futures XR Labs.
- Selected as a Ruhr Fellow for energy systems research in Germany, combining computer science expertise with energy economics and systems integration.
- Skilled in Python, PyTorch, OpenCV, Tableau, and business analytics tools.

Background or Justification

As UAV operations increase, ensuring the reliability and safety of motors and batteries has become a challenge. Traditional approaches for assessing motor and battery health, such as visual inspections or monitoring of only SoC and voltage, are often insufficient for autonomous UAV operations. Faults can occur during flight and may not be observable, while battery degradation trends are difficult to infer from simple telemetry. To address these challenges, the team proposes an integrated framework that combines mechanical design, sensor integration, and machine learning to enable real-time in-flight monitoring of powertrain health.

Brushless DC motors are commonly used in UAVs for their high reliability as a result of the lack of mechanical brushes and high-power density [3]. Due to their high speeds, they are subject to various faults such as bearing wear, coil degradation, and coupled propeller damage [3]. These faults can lead to increased vibration and decreased angular velocity and thrust [3]. Existing

methods such as visual inspections or manual preflight checks are labor intensive, subjective, and impractical for autonomous or remote UAV operations. Capturing motor sound with well-placed microphones, while minimizing mechanical vibration interference, is critical for fault detection. The team's experience with electromechanical systems and test instrumentation will directly inform the approach to microphone placement, vibration isolation, and test stand design.

Lithium-ion batteries degrade over time due to cycling and fatigue, high discharge rates, and high temperatures [4]. This results in reduced capacity, unstable voltage profiles, and shortened RUL [4]. Battery failures can cause midflight power loss, which is dangerous in UAV applications. Existing onboard telemetry provides only rough estimates of SoC, offering little insight into long term degradation or actual available flight endurance.

Lithium-ion battery health prediction requires high-quality data and predictive modeling. The team will leverage co-op and academic experience in data analytics, deep learning, and energy systems to understand degradation patterns, RUL estimation, and real-time monitoring. Supervised machine learning models, such as DeepONet, are well-suited for capturing complex relationships in voltage-current cycling data, enabling accurate prediction of battery health and remaining flight time.

Machine learning has emerged as a powerful approach for predictive maintenance, enabling early fault detection in safety-critical systems such as UAV powertrains [5]. Acoustic and vibro-acoustic signals are particularly well suited for monitoring rotating machinery, as they carry rich information about bearing wear, propeller damage, and other motor faults. Integrating these signals with 1D CNNs has shown strong performance in identifying early degradation patterns with high accuracy [6].

Autoencoder architectures have further advanced fault detection by enabling unsupervised feature extraction and anomaly detection without requiring extensive labeled fault datasets [7]. These machine learning models are well suited for UAV applications, where real-time monitoring and computational efficiency are essential. Recent work has shown that combining acoustic sensing with lightweight deep learning models allows effective fault detection directly on embedded platforms [8]. This integration of acoustic data and deep learning forms the basis for developing reliable, in-flight motor health monitoring systems for UAVs [9].

Building on the use of machine learning for motor fault detection, similar deep learning techniques have shown strong potential for modeling and predicting lithium-ion battery behavior, which is critical for ensuring safe drone operations. Unlike motors, where acoustic features provide direct fault signatures, battery systems require accurate modeling of voltage dynamics over time to enable early fault detection and energy management. Recurrent neural networks such as LSTM have been widely applied for this purpose, effectively capturing nonlinear temporal dependencies in voltage data [10], [11]. By learning from current and voltage time series, these models provide reliable short-term voltage forecasts, enabling timely intervention to prevent mid-flight failures.

More recent approaches have enhanced the predictive capabilities of these models by combining convolutional layers for feature extraction with temporal modeling components, leading to improved generalization under dynamic operational conditions [11]. Emerging operator learning

techniques such as DeepONet further address limitations of traditional sequence models by learning continuous mappings informed by physical principles, making them suitable for extrapolation across different battery types and use cases [12], [13]. This is especially valuable in UAV settings, where batteries may degrade differently across missions. Together with acoustic-based motor fault detection, these data-driven battery models form a complementary part of a larger health monitoring framework that enhances the reliability, safety, and autonomy of drone systems [14].

For in flight monitoring, machine learning models will be deployed on embedded computing platforms such as the NVIDIA Jetson. Previous software co-op and research experience in deploying deep learning models in real-time, integrating hardware and software components, and validating systems under operational conditions provides the team with the skills necessary to ensure successful implementation.

Statement of Work

The primary objective of this project is to develop a comprehensive in flight UAV powertrain health monitoring framework to ensure real time detection and prediction of faults in motors, propellers, and batteries. General objectives are listed below.

1. Design and assemble a UAV platform with integrated sensors, including microphones and electrical measurement hardware.
2. Develop a semi-supervised machine learning framework using a 1D CNN Autoencoder to detect anomalies in motor and propeller operation.
3. Implement a supervised DeepONet to predict battery degradation, estimate RUL, and compute remaining flight time.
4. Integrate the models onto onboard UAV computing hardware (NVIDIA Jetson) for real-time monitoring.
5. Validate the system through both stand-mounted and free-flight experiments to ensure reliability in realistic operational conditions.

The project will progress through the following phases. Phases may be circled back upon during the project and tasks may not be sequential. Many of the mechanical tasks will begin after initial data recording has been complete and used to optimize sensor type and placement. A more detailed task breakdown is found in Table 1.

The first phase involves mechanical platform design and sensor integration. During this phase, initial microphones will be chosen for use with the NVIDIA Jetson to record motor data. The microphones will be selected for optimal acoustic signal capture and frequency range. Other microphone types will be selected to perform a feasibility study on the preferred audio sensor. Once initial data is recorded using the preliminary microphones, vibration isolation will be designed to optimize the audio sensors. The second UAV will then be assembled with integrated electrical sensors for voltage and current measurements. Initial motor testing will begin using the stand mount configuration. For battery testing, a sensor system will be designed to approximate inner battery temperature. Finally, a strain gauge system will be designed for alerting models to structural stresses in the quadcopter arms.

In the second phase, motor acoustic signals will be recorded under healthy and induced fault conditions, and battery voltage-current data will be collected across multiple charge-discharge cycles. Research will guide the optimal microphone placement, appropriate levels of induced faults, and selection of degraded battery states for realistic testing. Experiments will be designed to control various conditions and scenarios, maximizing data quality for model training. Raw signals will be processed to remove noise, normalize readings, and extract features relevant to machine learning.

The third phase focuses on machine learning model development. The 1D CNN Autoencoder will be trained on healthy motor data to detect anomalies in propeller and motor operation. The DeepONet will be trained on battery cycling data to predict degradation, RUL, and remaining flight time. Models will be compared against baselines including logistic regression, supervised versus semi-supervised approaches, and LSTM networks versus DeepONet. Iterative refinement will ensure model accuracy, reliability, and computational efficiency, with particular attention to generalization across different batteries and flight conditions.

The fourth phase addresses software integration and in-flight testing. Trained models will be deployed on onboard NVIDIA Jetson hardware, with communication and data pipelines developed for real-time inference. Flight tests will evaluate system performance in detecting motor faults, monitoring battery health, and estimating remaining in-flight time based on UAV maneuvers. After each flight, flight logs and sensor data will be downloaded and analyzed to identify discrepancies, validate model predictions, and inform adjustments to mechanical configuration or model parameters.

The final phase involves analysis, validation, and reporting. Model outputs will be compared against sensor ground truth to validate performance across multiple scenarios. Analysis will include identification of failure modes, assessment of model accuracy, evaluation of real-time inference performance, and insights into sensor placement and mechanical design. Lessons learned will guide recommendations for UAV health monitoring frameworks. Final deliverables include comprehensive reports, presentations, and demonstrations for faculty, advisors, and project stakeholders, documenting the transition from theoretical feasibility to practical in-flight monitoring.

The methods used in this project fall into three categories: mechanical, software, and testing. For mechanical, methods used includes UAV assembly, sensor engineering and design, sensor mounting and vibration isolation, and stand mounted and in-flight testing. For software, methods used includes Python-based data preprocessing, deep learning model development (1D CNN Autoencoder and DeepONet), feature extraction, model training, validation, and onboard deployment. For testing, methods used includes induced fault scenarios on stand-mounted UAV, real-world free-flight validation, cross-verification with sensor measurements.

This project will have several deliverables, which are shown below.

1. Fully assembled and instrumented UAV platform.
2. Datasets for motors and batteries using stand mounted and in-flight testing.
3. Trained machine learning models for motor anomaly detection and battery degradation prediction.

4. Integrated system capable of running on onboard UAV hardware for real-time monitoring.
5. Final report summarizing methodology, experiments, results, and recommendations.
6. Presentation materials for faculty advisors and client demonstration at senior expo.

While this project's goal is to produce a monitoring framework, the initial data collection will be limited to the specific quadcopter used for testing. Both the motor and battery models are expected to be applicable to other quadcopter and octocopter builds; however, verifying this is beyond the current scope. The health framework will be configured to run on a NVIDIA Jetson, but further in-flight computing methods will not be considered. Additionally, this work will compare multiple types of microphones/audio sensors, though further specifications regarding specific sensor types will not be addressed. The health monitoring of non-powertrain subsystems and large-scale deployment or fleet management beyond experimental testing and validation will not be considered in this project. Finally, UAV navigation, flight control, and communication system development will not be considered for this project.

Schedule and Management Plan

This project consists primarily of testing, software development and integration, and mechanical design. Data collection for the propeller/motor and battery models is required before any model development can begin. Once initial data is acquired, work can start on developing and integrating the models onto the NVIDIA Jetson. The mechanical tasks will operate largely independent of the software development until they are integrated into the final models and the outfitted drone.

Major milestones such as project reviews will be started at least one week beforehand. Anticipated decision points/design freezes have been accounted for in Table 1. This project will require little to no fabrication that requires machining or processing. The major tasks of the project have been broken down into smaller components and displayed graphically in Figure 1.

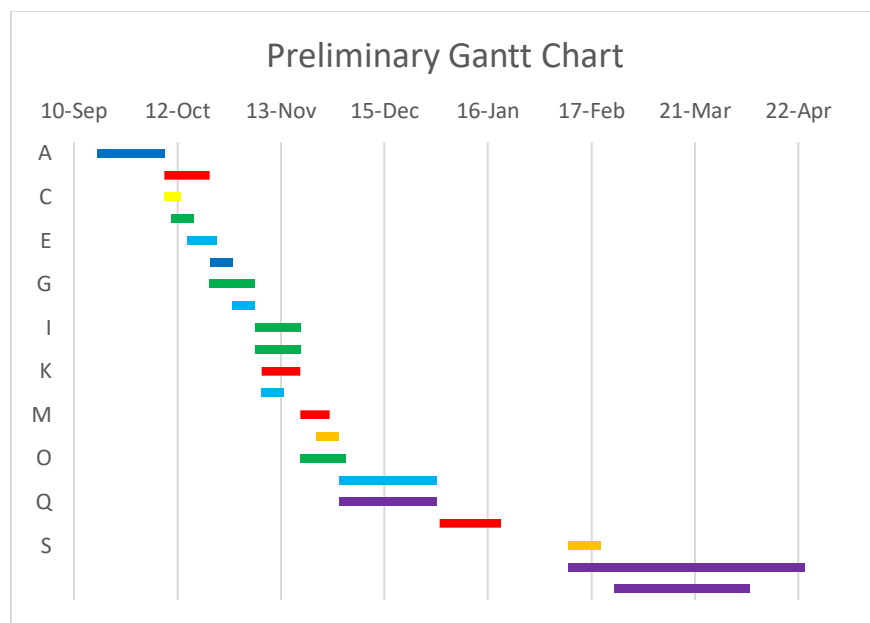


Figure 1: Preliminary Gantt Chart

Table 1: List of Tasks

	Task	Start Date	End Date	Classification	Code
A	Record motor and propeller data	17-Sep	8-Oct	Testing	
B	Create initial motor model	8-Oct	22-Oct	Machine Learning	
C	Decision making on batteries	8-Oct	13-Oct	Other	
D	Research into stress and temperatures sensors	10-Oct	17-Oct	Mechanical	
E	Record round one battery data	15-Oct	24-Oct	Testing	
F	Integrating in code to Jetson for in flight	22-Oct	29-Oct	Hardware Integration	
G	Build second drone	22-Oct	5-Nov	Mechanical	
H	In-flight motor testing	29-Oct	5-Nov	Testing	
I	Research and start design of vibration isolation	5-Nov	19-Nov	Mechanical	
J	Integrate and test second microphone array	5-Nov	19-Nov	Mechanical	
K	Finetune motor model	7-Nov	19-Nov	Machine Learning	
L	Round two battery data	7-Nov	14-Nov	Testing	
M	Create initial battery model	19-Nov	28-Nov	Machine Learning	
N	Prepare for 1st Design Review	24-Nov	1-Dec	Review Preparation	
O	Integrate strain gauges on second drone	19-Nov	3-Dec	Mechanical	
P	Further battery data collection	1-Dec	31-Dec	Testing	
Q	Motor Documentation	1-Dec	31-Dec	Documentation	
R	Test and finetune battery model	1-Jan	20-Jan	Machine Learning	
S	Prepare for 2nd Design Review	10-Feb	20-Feb	Review Preparation	
T	Write Final report	10-Feb	24-Apr	Documentation	
U	Expo	24-Feb	7-Apr	Documentation	

Costs

The project's cost will be primarily material based. Many of the required materials are already available and will therefore not be included in the budget. The drone to be used is already built, and the parts are available to construct a second for further sensor testing. Lightly used propellers and BLDC motors, which are assumed to be faulty, will also be provided. Sensors to measure component noise and mechanical stress will be purchased, along with general electrical components such as breadboards and wires. Further sensors may be purchased if determined necessary. Decision making regarding battery data collection has not occurred, so materials needed for that phase are unclear, but a temperature sensor will likely be needed. All software required will be accessible to students on campus. The price breakdown per material and total estimated cost is shown below in Table 2, and student hours are not considered within the budget.

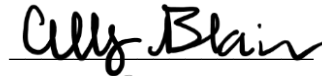
Table 2: Preliminary Material Cost Breakdown

Material(s) Required	Estimated Cost
USB microphones	\$22
MEMS I2S microphones	\$12
Breadboards and associated wires	\$10
Strain gauges	\$40
Thermocouple/thermistor and equipment	\$40
Additional sensors	TBD
Lightly used propellers and BLDC motors	N/A
Components to build second UAV	N/A
NVIDIA Jetson Nano	N/A
All materials	\$124

Submission:

The engineers and computer scientists of Team W02 respectively submit this proposal for your consideration.

Ally Blair



Prishha Chawla

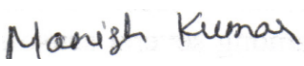


Siddarth Urankar



Your signature below acknowledges your agreement to the scope of work, proposed timeline, and anticipated cost. Further, it acknowledges that this is a student design project and understand that this is a learning experience.

Client Representative Name: Manish Kumar

Signature: 

Date: 10/15/2025

Works Cited

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